Inorganic CUME on Materials Chemistry
November 3rd, 2017
Answer Any Five. Only Five Will Be Graded

1. (20 points) Briefly explain the working principle of X-ray diffraction (XRD). What is the Bragg's Law? How to use Bragg's Law to calculate the lattice constants in a cubic structure? Calculate the number of atoms in each of the four different unit cells below (Figure 1).

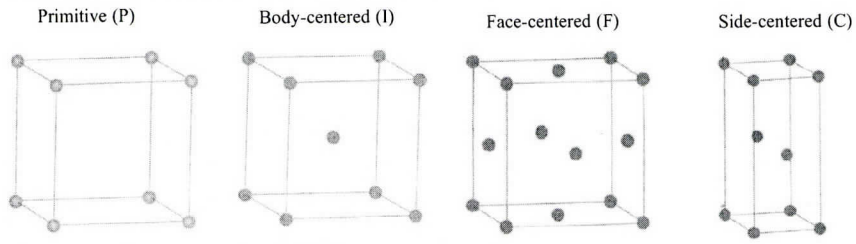

Figure 1. Four types of different unit cells

2. (20 points) What is Nuclear Magnetic Resonance (NMR)? What is Electron Paramagnetic Resonance (EPR)? Briefly discuss similarities and differences between NMR and EPR? Give an example of using NMR or EPR for chemistry research.

3. (20 points) Give the coordination number and charge of the central metal ion in the following coordination compounds: **(a)** $Na_2[Zn(OH)_4]$; **(b)** $K[Co(H_2O)_2(C_2O_4)_2]$; **(c)** $[Ru(H_2O)_2(NH_3)_2Cl_2]Br$.

4. (20 points) Iron(II) forms a complex in hemoglobin. For each of the two octahedral complex ions $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$, draw an energy diagram showing orbital splitting, predict the number of unpaired electrons, and identify the ion as low spin or high spin.

5. (20 points) Prussian blue can be made by the addition of Fe(II) to $K_3Fe(CN)_6$ or Fe(III) to $K_4Fe(CN)_6$. All of these reagents are weakly colored. Prussian blue however is an intense blue color and one of the earliest synthetic pigments. Describe the reaction responsible for the formation of the solid called Prussian Blue, describe the structure of the solid and describe the electronic transition responsible for the blue color.

6. (20 points) Write spontaneous redox reactions and rank oxidizing and reducing Agents by strength. (a) Combine the following three half-reactions into three balanced equations for spontaneous reactions (A, B, and C), and calculate $E°_{cell}$ for each; You will only need to use two half-reactions for each balanced equation; (b) Rank the relative strengths of the oxidizing and reducing agents

(1) $NO_3^-(aq) + 4H^+(aq) + 3e^- \rightarrow NO(g) + 2H_2O(l)$ $E° = 0.96$ V

(2) $N_2(g) + 5H^+(aq) + 4e^- \rightarrow N_2H_5^+(aq)$ $E° = -0.23$ V

(3) $MnO_2(s) + 4H^+(aq) + 2e^- \rightarrow Mn^{2+}(aq) + 2H_2O(l)$ $E° = 1.23$ V

7. (20 points) In one or two short sentences each, explain what is meant by the following terms. a) galvanic or voltaic cell; b) electrolytic cell; c) salt bridge; d) secondary battery or cell; e) primary battery or cell; f) glass electrode

8. (20 points) Why diamond is electrically insulating and hard, while graphite is electrically conductive and slippery, although they are both made of pure carbon? Which one, diamond or graphite, is the most stable form at standard thermodynamic conditions? What is the difference between quartz and glass in structural chemistry?

9. (20 points) Lithium is a very active metal. Does Li have a body-centered or face-centered cubic structure? Lithium and its compounds have several industrial applications. Give an example for those applications. A copper foil substrate (about 8 μm thick) was coated with a very thin layer (nanometer range) of lithium. Please list at least three different approaches to measure the thickness of coated lithium.

10. (20 points) The chemical potential of interstitial Li in an electrode material is equal to the derivative of the free energy of the material with respect to Li concentration according to

$$\mu_{Li} = \frac{\partial g}{\partial x}$$

Here g is the Gibbs free energy per LixM formula unit, where M represents the host chemistry (e.g., Si, CoO_2 etc) and x denotes the fraction of available sites occupied by Li. Graphically, the Li chemical potential within a compound at a particular composition x is equal to the slope of the free energy g at that composition as illustrated in the Figures 1a-b below. The straightforward connection between a measurable voltage curve and the Gibbs free energy means that the occurrence and nature of phase transformations due to variations in Li concentration will have clear signatures in the voltage profile. For example, if Si forms a solid solution with Li as in Li_xSi during lithiation process (see Figure 1a below, assume Li concentration is up to 1 during the reaction), it will exhibit a smooth sloping voltage curve as illustrated in Figure 1b below. If the lithiation process of Si is accompanied by a first-order phase transformation from a Li poor phase, **α**, to a Li rich phase, β (Figure 1c), predict the corresponding voltage curve and explain why.

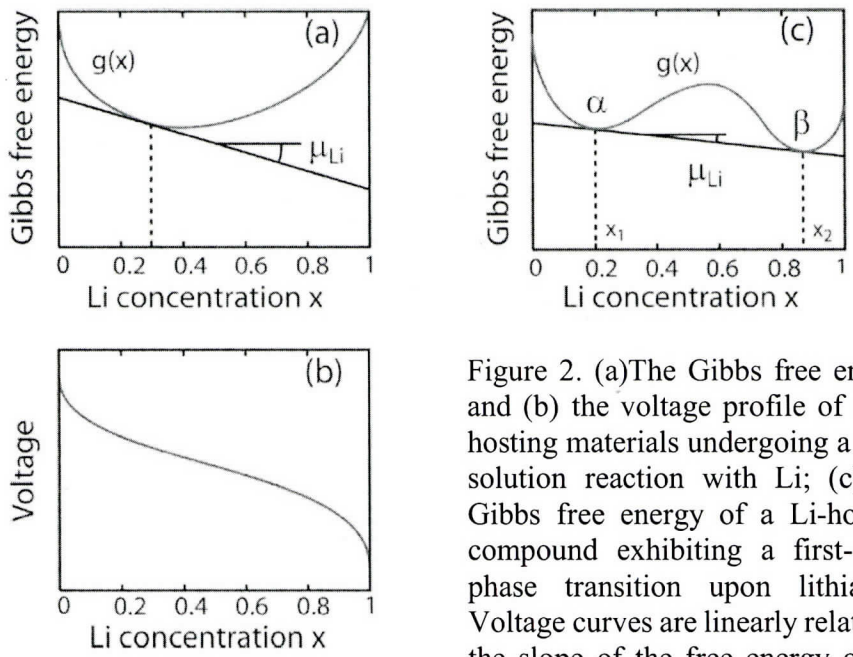

Figure 2. (a)The Gibbs free energy and (b) the voltage profile of a Li-hosting materials undergoing a solid solution reaction with Li; (c) the Gibbs free energy of a Li-hosting compound exhibiting a first-order phase transition upon lithiation. Voltage curves are linearly related to the slope of the free energy of the electrode material.